

Project Proposal

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Problem

Organizations routinely need to share database views with analysts and partners while keeping privacy risks low. In practice, this sharing is often implemented with hand-written SQL and ad-hoc masking rules that are easy to misapply and difficult to audit. This project proposes a model-driven engineering (MDE) toolchain that accepts (i) a relational schema annotated with sensitivity information and (ii) a privacy-policy model, then automatically generates privacy-preserving SQL views together with a concise sharing report. The approach is aligned with the GDPR's definition of *pseudonymisation*—processing that prevents attribution without separately held “additional information” [1]—and reflects recent EU guidance on selecting effective techniques and safeguards, supporting a pragmatic “privacy by design” control [2].

Models and Metamodels

The project spans two domains: data modeling and privacy/governance. Figure 1 presents the overall structure and the metamodels used. Four Ecore metamodels structure the solution.

SchemaMM describes the shareable dataset with classes for **Table**, **Column** (including **type**, **category: PIIType**, and **sensitivity**), **ForeignKey**, and optional **DerivedColumn(expr)**. **PolicyMM** encodes pseudonymisation policy as **Rules** over either specific columns or PII categories, each rule selecting a technique (e.g., **hash**, **tokenize**, **redact**, **partial**, **truncate**, **nullify**), with an optional condition, precedence, and rationale. The technique set and its attacker-aware use are grounded in ENISA's taxonomy and ISO/IEC 20889 terminology [3], [4]. **SQLViewMM** provides a platform-independent representation of generated database views, including **View**, **Projection** (annotated with **masked:boolean** and **technique**), **Join**, and **Filter**, with traceability from each **Projection** back to its source **Column**. **ReportMM** captures audit-friendly summaries—**Exposure(table, column, technique, legal_basis, purpose)**, **ResidualRisk**, and **RuleApplication**—inspired by the process framing in NISTIR 8053 [5].

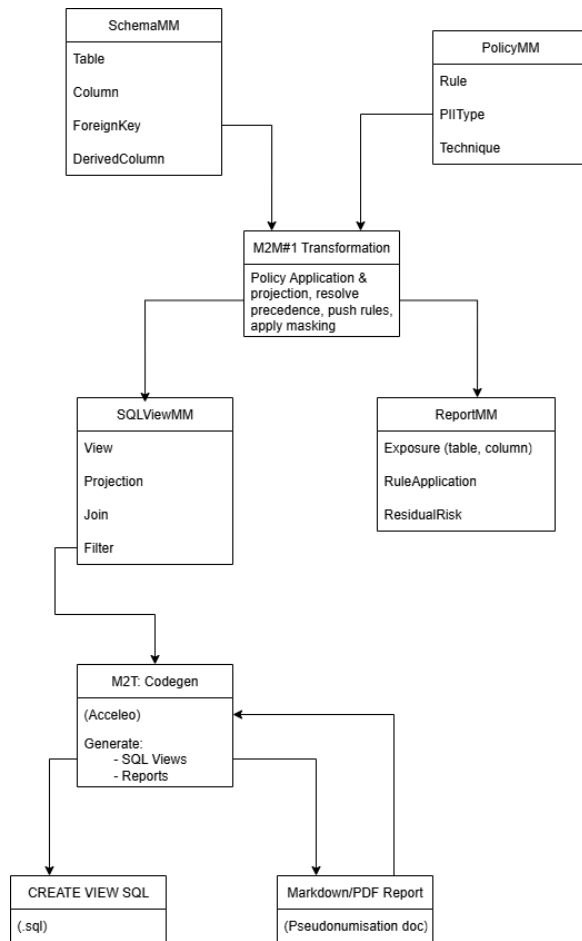


Figure 1

These metamodels are linked by a simple composition: **SchemaMM** combined with **PolicyMM** is transformed into **SQLViewMM** and **ReportMM**; those are then rendered into executable SQL (**.sql**) and human-readable documentation (**.md/.pdf**). Model validation is expressed in OCL to support accountability: every masked projection must declare a masking technique; no two rules may target the same object at the same precedence; and all join columns must exist and respect foreign-key constraints. Together, these checks help demonstrate appropriate technical and organizational measures under the GDPR [1].

Tools and Transformations

Modeling and editing are performed in Eclipse Modeling Tools with EMF/Ecore. Schema and policy can be authored either in a Sirius diagram editor or via a lightweight Xtext DSL, depending on the audience. The transformation pipeline comprises two M2M steps and one M2T step. First, **M2M#1** (ATL or QVTo) resolves policy precedence and conditions, propagates masking decisions into **Projections**, and pushes masking through simple **DerivedColumns** to

produce **SQLViewMM**. Second, **M2M#2** synthesizes **ReportMM** by summarizing which columns were masked, on what grounds, and with which residual risks or assumptions. Finally, **M2T** (Acceleo) emits portable **CREATE VIEW** scripts from **SQLViewMM** and a succinct Markdown/PDF report from **ReportMM**, using ISO 20889 vocabulary for clarity and consistency [4].

External tools both motivate and consume the outputs. In PostgreSQL, views provide a clean, re-evaluated sharing boundary that avoids materialized copies [7]. As additional evidence of feasibility, Microsoft's Dynamic Data Masking shows masking as a first-class database control in mainstream platforms; our project models such policies vendor-neutrally and generates portable SQL views independent of proprietary features [8]. Day-to-day evolution is straightforward: data owners adjust **Schema.model** or **PrivacyPolicy.model** and re-run the pipeline to regenerate views and reports. Trace links from **Column** → **Projection** → SQL enable selective regeneration and auditing. The scope is intentionally limited to pseudonymisation rather than full anonymisation (the latter being irreversible and out of scope here), consistent with regulator guidance [6]. Classic release-oriented techniques such as k-anonymity are noted for context but are less suitable for live, query-time sharing via SQL views.

References

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- [7] PostgreSQL Global Development Group, “CREATE VIEW,” *PostgreSQL 18 Documentation*, 2025. [Online]. Available: postgresql.org/docs/current/sql-createview.html. Accessed: Oct. 6, 2025.
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